

VIKULOVO, Russian Federation

WMO: 284810

Lat: 56.817N

Lon: 70.617E

Elev: 257

StdP: 14.56

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-28.3	-23.5	-33.7	0.8	-27.9	-29.0	1.1	-23.2	16.1	12.8	14.4	13.2	3.4	180	0.602

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.1	85.1	67.5	82.0	66.1	79.0	64.5	70.1	80.6	68.3	78.1	66.6	75.8	6.7	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.4	98.1	75.2	64.7	92.4	73.3	63.0	86.9	71.7	34.2	80.9	32.8	78.0	31.4	75.9	77.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.0	14.0	12.6	DB	-32.9	90.8	6.5	4.1	-37.6	93.7	-41.4	96.2	-45.1	98.5	-49.8	101.5	(4)
(5)			WB	-33.1	72.8	6.5	2.1	-37.8	74.3	-41.6	75.5	-45.3	76.6	-50.0	78.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	35.4	0.2	5.8	21.1	38.5	52.9	62.8	65.1	60.6	50.3	37.8	20.2	7.6		
	DBStd	24.97	13.74	13.14	10.11	10.21	9.94	8.49	6.71	7.14	8.04	9.10	12.75	14.43		
	HDD50	6835	1545	1238	895	366	87	9	0	5	95	388	894	1313		
	HDD65	11024	2010	1658	1360	796	391	139	84	175	445	844	1344	1778		
	CDD50	1505	0	0	0	20	177	393	470	333	103	9	0	0		
	CDD65	220	0	0	0	0	16	74	88	39	3	0	0	0		
	CDH74	2068	0	0	0	5	241	736	744	309	33	0	0	0		
	CDH80	496	0	0	0	0	54	216	163	60	3	0	0	0		
(14) Wind	WSAvg	6.1	5.5	5.9	6.8	7.0	6.8	5.8	5.2	5.3	5.8	6.2	6.4	6.2		
(15) Precipitation	PrecAvg	18.2	0.7	0.5	0.7	1.1	1.5	2.4	3.0	2.7	2.0	1.4	1.3	1.1		
	PrecMax	22.8	1.6	1.3	1.7	2.1	3.1	6.1	7.6	5.6	4.2	3.4	2.1	2.1		
	PrecMin	13.0	0.1	0.0	0.0	0.0	0.2	0.5	0.8	1.0	0.7	0.1	0.3	0.4		
	PrecStd	2.4	0.4	0.3	0.4	0.6	0.7	1.4	1.7	1.1	0.8	0.7	0.5	0.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.7	35.9	46.1	72.6	85.3	89.3	87.7	85.9	78.4	64.8	44.8	36.0		
		MCWB	31.4	32.3	38.3	55.5	60.9	68.1	69.8	69.0	60.9	52.0	41.1	33.9		
	2%	DB	28.3	31.9	40.4	65.2	80.1	85.8	84.3	81.0	72.4	59.1	40.6	33.1		
		MCWB	27.0	29.7	35.0	51.3	59.5	67.1	68.7	66.8	59.2	49.3	38.2	31.5		
	5%	DB	24.4	28.5	37.2	59.3	76.2	82.0	81.3	77.3	67.3	54.9	37.9	29.7		
		MCWB	23.1	26.8	33.2	48.5	57.6	65.9	67.4	65.1	56.9	47.3	35.9	28.3		
	10%	DB	20.4	24.5	34.7	54.3	71.1	77.9	78.3	73.7	63.4	51.0	35.0	26.4		
		MCWB	19.4	22.8	31.5	45.5	55.3	64.0	66.1	63.4	54.3	45.2	33.2	25.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	31.7	32.7	38.8	57.2	64.5	72.0	72.6	70.8	63.5	53.8	41.7	34.4		
		MCDB	32.6	35.3	44.6	67.5	78.7	84.3	84.0	81.8	74.3	61.8	43.9	35.6		
	2%	WB	27.2	29.7	35.7	52.2	61.3	69.3	70.6	68.4	60.2	50.8	38.7	31.5		
		MCDB	28.3	31.7	39.2	63.4	76.5	80.8	80.7	77.5	70.2	57.2	40.5	33.1		
	5%	WB	23.2	26.9	33.8	49.2	59.0	67.2	69.0	66.5	57.7	48.2	35.9	28.5		
		MCDB	24.5	28.6	36.8	58.9	73.4	78.2	78.6	75.2	66.0	53.9	37.7	29.7		
	10%	WB	19.3	22.9	31.7	46.0	56.5	65.2	67.2	64.6	55.2	45.4	33.4	25.1		
		MCDB	20.5	24.3	34.6	53.7	69.0	75.8	76.0	71.7	62.0	50.5	34.8	26.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	12.5	15.3	15.7	16.2	20.3	18.4	17.1	17.0	15.8	12.1	10.6	12.4		
		MCDBR	12.8	14.5	15.0	24.1	27.1	24.0	21.6	22.5	22.6	19.6	9.4	13.7		
	5% WB	MCWBR	12.4	13.4	10.9	13.5	12.0	9.9	9.3	10.3	11.4	11.7	8.3	12.8		
		MCWBR	12.5	14.5	13.9	22.6	24.8	21.4	19.6	20.3	21.2	17.6	9.4	13.8		
(40) Clear-Sky Solar Irradiance	taub		0.277	0.311	0.368	0.413	0.407	0.409	0.425	0.429	0.377	0.353	0.311	0.279		
	taud		2.035	2.019	1.934	2.051	2.229	2.288	2.242	2.198	2.305	2.220	2.087	1.979		
	Ebn at Noon		180	225	240	251	263	264	256	245	242	210	163	139		
	Edh at Noon		19	30	43	45	40	39	40	39	30	25	19	16		
(44) All-Sky Solar Radiation	RadAvg		243	567	1049	1448	1687	1802	1713	1319	876	444	208	148		
	RadStd		19	38	74	113	128	146	153	119	95	60	36	19		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page